Venkata Lakshmi Parimala Pasupuleti

Forward Deployed Engineer · AI · Python · Full-Stack · Claude Code · LLM Integration · APIs & Webhooks · Rapid Prototyping · Enterprise Security & Governance · End-to-End Delivery

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PROFESSIONAL SUMMARY

Forward-deployed engineer with 8 years of end-to-end production delivery - and two AI systems that demonstrate what "embed, scope, build, and ship in days, not quarters" looks like in practice. LeaseI is a three-surface enterprise AI platform built solo from discovery through deployment: document intelligence, 8-agent LangGraph support investigation, and a tenant self-service layer - scoped independently, prototyped rapidly with Claude Code and AI coding tools, shipped to production with full testing and handoff documentation. AgentProbe is an open-source LLM evaluation framework built the same way: 15-endpoint FastAPI service, CI-integrated regression gate, multi-provider routing - zero regressions shipped across 1,240+ test cases. At Entrata, delivered 20+ production modules across 120+ Agile sprints, owning requirements, architecture, testing, deployment, and stakeholder communication end-to-end. Security and data governance are built-in, not bolted on: 100% PHI compliance with Libsodium-encrypted pipelines and zero security incidents across 7 years. Daily Claude Code user; Anthropic API integrated throughout. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & LLM Tools: Claude Code, LangChain, LangGraph, Agentic Workflows, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, HITL, Guardrails, Structured Outputs, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Anthropic API, OpenAI API, Ollama, MCP, OCR, BERT, spaCy, NLP, PyTorch, scikit-learn

Full-Stack & Backend: Python (8yr), TypeScript, JavaScript, React.js, FastAPI, Flask, REST APIs, Webhooks & API Integration, Frontend Development, Database Design, PostgreSQL, MySQL, Oracle, Redis, NoSQL, Docker, Deployment & CI/CD (GitHub Actions), Testing (Unit, Integration, End-to-End), AWS (S3, EC2, RDS, Glue), RabbitMQ, Microservices, Linux, Git

Security, Governance & Delivery: Enterprise Security & Compliance, Data Governance, PHI Compliance (Libsodium), Secure System Design, HIPAA-Compliant Architecture, Rapid Prototyping, End-to-End Project Ownership, Cross-Functional Stakeholder Communication, Technical Documentation, Observability, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern - Forward Deployed

- Embedded with a business stakeholder team to scope, prototype, and deploy a production agentic AI system (LangGraph, FastAPI, Python) end-to-end - used Claude Code and AI coding assistants throughout to compress build cycles; delivered working software in weeks, not quarters.
- Owned architecture, testing, deployment, and stakeholder handoff; built observability on agent performance and retrieval quality; codified reusable patterns from the engagement for future deployments.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, APIs, and software engineering; designed and deployed observability dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **End-to-end full-stack delivery across 120+ Agile sprints** - 20+ production Python and JavaScript/TypeScript modules shipped from requirements through deployment; owned architecture decisions, testing, stakeholder communication, and rollout for each engagement; 99.5% uptime SLA at 100K+ monthly active users; Docker containerization; AWS S3, EC2, RDS, Glue; webhook and API integrations with external data providers.
- **Enterprise security, data governance, and compliance** - 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents across 7 years); designed data governance controls for HUD income-limit pipeline (180,000 domain records powering housing eligibility decisions); HIPAA-adjacent data handling; Redis caching; 200+ SQL query optimizations (30% performance gain); AI-driven observability reducing triage time 40%.
- **Rapid prototyping, stakeholder communication, and engineering leadership** - translated ambiguous requirements from product, compliance, and operations stakeholders into scoped, working software; 300+ code reviews; maintained sub-10 bug count for 6+ consecutive months; mentored 5+ engineers; leveraged Claude Code and AI development tools to accelerate prototyping, debugging, and documentation.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: End-to-End AI Platform - Solo Discovery through Deployment

FDE · Full-Stack · Agentic

Python · LangGraph (8-agent) · FastAPI · React · TF-IDF · OCR · SQLite · Streamlit | github.com/venkatalakshmiparimala/Lease_complaince_raq

Three-surface enterprise AI platform built solo from scope through production - document intelligence (OCR, section-aware chunking, 100% citation correctness), 8-agent LangGraph support investigation (tool routing, memory, HITL escalation, source-linked root-cause hypotheses), and tenant self-service with deterministic governance guardrails. Rapid prototyping with Claude Code throughout. 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

AgentProbe: LLM Evaluation & CI Regression Gate

Evals · Claude Code · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform built end-to-end with Claude Code: 15-endpoint FastAPI REST service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, multi-provider routing for cost optimization. Codified reusable evaluation patterns - zero regressions shipped across 1,240+ test cases. Open-source.

AI on FHIR: Full-Stack NLP System with HIPAA Compliance

Full-Stack · Security · APIs

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs · FHIR | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application (Python backend + React.js/TypeScript frontend) translating natural-language queries into FHIR REST API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, data governance, and full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017